

MERN Stack Wind Data Platform – Short Pipeline & 15-Day Task Plan

1. Project Objective

Develop a MERN stack application for uploading wind data CSV files, performing generalized schema validation, quality control validation, wind analytics processing, and real-time dashboard visualization.

2. Complete System Pipeline

CSV Upload (Frontend)

- CSV Parsing (PapaParse)
- Dynamic Column Mapping
- Generalized Schema Validation
- Quality Control Validation
- Valid Data Storage (MongoDB)
- Invalid Data Storage (Error Logs)
- Analytics Processing
 - Missing Data Detection
 - Turbulence Intensity (TI)
 - Shear Analysis
 - Weibull Fit
- WebSocket Live Streaming
- Frontend Dashboard Visualization

3. Validation Rules

- Missing or NaN wind speed/wind direction values
- Wind speed outside 2–60 m/s
- Wind direction outside 0–360°
- Five consecutive identical wind speed/direction values
- Invalid timestamp or datatype
- Missing required columns

4. Generalized Schema Validation

Different CSV formats are supported using a dynamic column mapping engine. Alias dictionaries and header normalization automatically map vendor-specific column names to a common internal schema.

5. Main Modules

1. CSV Upload & Parsing
2. Dynamic Column Mapping
3. Schema Validation

4. Quality Control Validation
5. Error Logging
6. MongoDB Storage
7. Wind Analytics Engine
8. WebSocket Streaming
9. Frontend Dashboard

6. 15-Day Timeline

Timeline	Tasks
Day 1-2	Project setup and MongoDB schema design
Day 3-4	CSV upload API and column mapping engine
Day 5-6	Schema validation and QC validation
Day 7	Error logging system
Day 8	MongoDB optimization
Day 9	WebSocket integration
Day 10-11	Frontend charts and error dashboard
Day 12	TI and Shear analysis
Day 13	Weibull fitting and KPI dashboard
Day 14	Final dashboard integration
Day 15	Testing and documentation

7. Expected Deliverables

- Working MERN stack application
- CSV upload and validation pipeline
- Error logging dashboard
- Wind analytics visualizations
- Real-time WebSocket updates
- Documentation and demo